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RESEARCH ARTICLE

Upcycling of Polycaprolactone via Nonthermal Plasma-Enhanced Catalytic In Situ Hydrogenolysis at Ambient Pressure

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ABSTRACT

Chemical recycling of plastics is typically hampered by energy-intensive processes requiring harsh conditions. Here, we report an electrified non-thermal plasma (NTP) catalysis strategy for the selective hydrogenolysis of polycaprolactone (PCL) into valuable caprolactones at ambient pressure and without solvents. By replacing high-pressure H_2 and noble-metal catalysts with an atmospheric methane plasma over a Ni/HY catalyst, this approach achieves complete PCL conversion and a remarkable 93.1% yield of caprolactones at 150°C utilizing solely plasma-generated heat. In situ hydrogen radicals provided by the plasma and Ni sites regenerate Brønsted acid sites, driving the selective dissociation of the PCL alkoxy bond and enabling the preferential formation of γ-caprolactone (γ-CL) both kinetically and thermodynamically at low temperatures. Simultaneously, plasma polarization promotes electron redistribution between HY and PCL, significantly lowering the dissociation energy barrier of PCL. Compared with thermal catalysis at 200°C, synergistic plasma catalysis reduces the PCL activation energy barrier from 2.29 to 1.71 eV and increases the PCL conversion by 8.77-fold. We further demonstrate the scalability of this process by operating a 100 g batch system powered by photovoltaics. This work establishes a synergistic pathway integrating methane cracking with waste plastic upcycling, offering a practical route for closed-loop polymer upcycling.

1 | Introduction

The linear produce-use-discard paradigm governing global plastic consumption has precipitated an environmental crisis, necessitating an urgent transition towards a circular economy [1–4]. In this context, polyesters exemplify the sustainability challenge, driven by their massive global production and ubiquitous usage [5, 6]. Unlike mechanical recycling constrained by material degradation, chemical recycling provides a transformative route to closed-loop sustainability [7]. By targeting the selective cleavage

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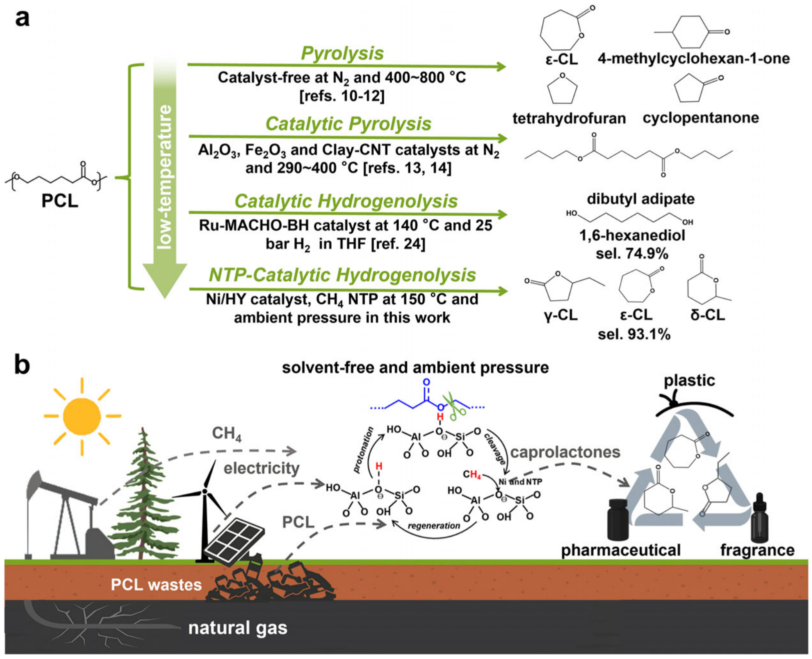

FIGURE 1 | (a) Product distribution and reaction conditions of PCL depolymerization via pyrolysis and other chemical depolymerization approaches; (b) schematic diagram of the plasma-catalytic process for solvent-free hydrogenolysis of PCL into CLs, powered by clean energy and natural gas as the plasma source gas.

of ester linkages to recover virgin-quality monomers or value-added chemicals, this strategy effectively enables the resource valorization of waste polyester plastics.

Among polyesters, biodegradable polycaprolactone (PCL) stands out for its exceptional biocompatibility and superior permeation properties, finding extensive applications in tissue engineering, drug delivery systems, and agricultural films [8, 9]. However, relying solely on natural biodegradation for end-of-life management represents a significant economic loss, given that the resulting degradation products are chemically irrecoverable and lack economic value. Consequently, the chemical recycling of PCL is particularly compelling, as it enables a resource-efficient cycle that simultaneously addresses waste management and recovers high-value feedstocks. While conventional thermochemical methods, such as pyrolysis, allow for PCL depolymerization under solvent-free and ambient-pressure conditions, they typically suffer from high energy consumption and low selectivity, resulting in complex product mixtures [10]. As shown in Figure 1a, pyrolysis of PCL above 400°C yields a mixture with a variety of valuable compounds, e.g., dibutyl adipate, cyclopentanone, and 1,6-dioxacyclododecane-1,12-dione, but the yield of individual products remains relatively low due to the temperature-sensitive intermediates that may undergo side reactions [11, 12]. Although catalysts (e.g., Fe_2O_3, Al_2O_3 [13], CNT-clay [14], $MgCl_2$ [15]) can lower energy barriers and promote lactone formation, achieving high selectivity for the strained and valuable γ-caprolactone (γ-CL) is particularly challenging. This is constrained by both thermodynamic and kinetic limitations; while the more stable ε-caprolactone (ε-CL) can be obtained under milder conditions [16, 17], the formation of γ-CL necessitates elevated temperatures to drive the requisite skeletal rearrangement of PCL-derived fragments in conventional thermal approaches [15].

Non-thermal plasma (NTP) catalysis has recently emerged as an innovative external-field-assisted thermal depolymerization process to circumvent these thermodynamic constraints in plastic depolymerization [18, 19]. This nonthermal process utilizes energetic electrons to generate a rich flux of reactive radicals and species from an ionized source gas, enabling synergistic reactions on catalyst and reducing the thermal energy input required for depolymerization. For instance, N_2 plasma combined with ZSM-5 or $NiCeO_x/\beta$ catalysts has been shown to pyrolyze HDPE into H_2 and light hydrocarbons with much higher yields than conventional catalytic pyrolysis [20–22], while H_2 plasma enabled > 70% selective C_2H_4 production from PS at 400°C [10, 23]. In the context of hydrogenolysis—a process traditionally reliant on noble metals (e.g., Ru, Pt) and high-pressure H_2 for H_2 cleavage—plasma catalysis emerges as a promising alternative [24–29]. It provides a promising pathway to generate reactive hydrogen species under mild conditions, which subsequently participate in situ in the catalytic depolymerization of plastics. This approach effectively eliminates the dual requirement for precious-metal catalysts and high-pressure H_2 supplies inherent to conventional thermal hydrogenolysis. Furthermore, as an electrified technology, the plasma process can be powered by decentralized, intermittent renewable electricity (e.g., solar, wind), enabling plastic depolymerization processes with the potential for a significantly reduced carbon footprint.

Herein, we report a sustainable strategy that synergistically integrates methane cracking with plastic waste upcycling (Figure 1b). By utilizing an earth-abundant CH_4 plasma over a non-noble Ni/HY catalyst at ambient pressure in a packed-bed dielectric barrier discharge (DBD) reactor (Figure S1), we completely circumvent the cost-prohibitive and carbon-intensive reliance on molecular H_2. Mechanistically, Ni sites promote CH_4 disso-

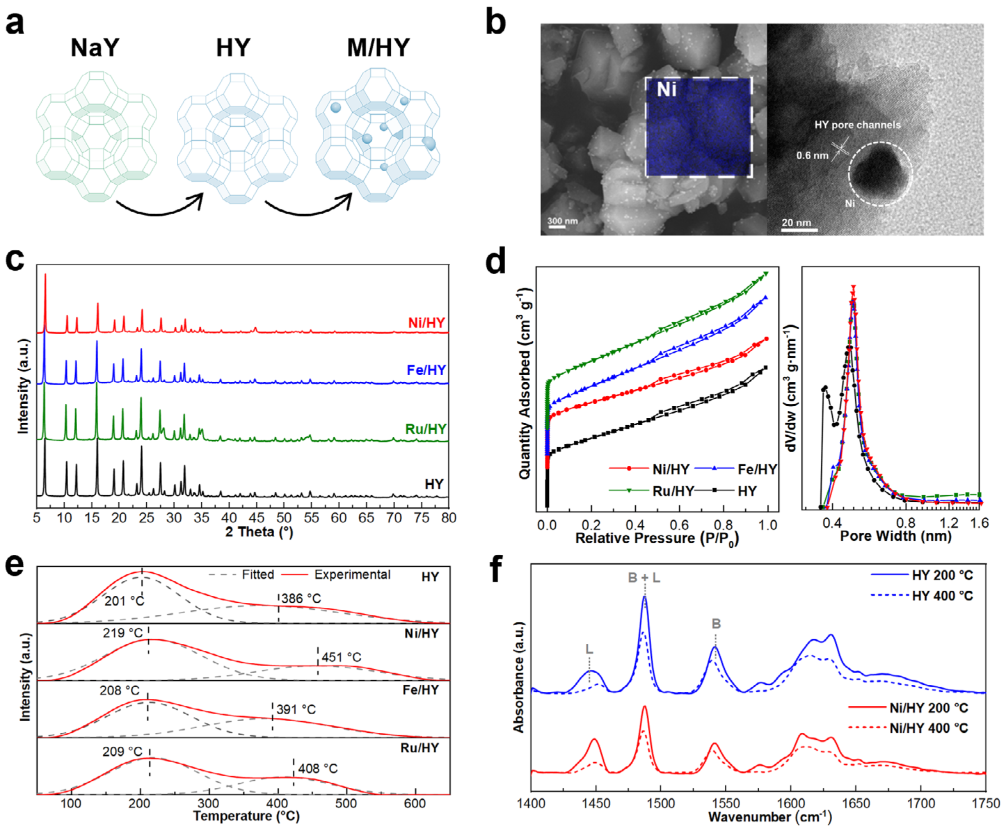

FIGURE 2 | Synthesis and physicochemical properties of M/HY catalyst. (a) Schematic illustrates the preparation process of HY and metal-loaded M/HY catalysts (taking Ni/HY as an example). First, NaY was synthesized and modified via ion exchange to obtain HY, followed by loading the active metal onto HY through the impregnation method to synthesize M/HY. (b) SEM-EDS and HRTEM images of Ni/HY. (c) XRD patterns of M/HY match PDF #43-0618, representing its FAU structure. d N_2 physisorption-desorption isotherms and pore size distribution. (e) NH_3-TPD patterns of M/HY represent the acid density when doping different active metals. (f) Py-FTIR patterns of HY before and after Ni loading represent the ratio of Lewis (labeled as L) and Brønsted (labeled as B) acid density.

ciation, supplying in situ hydrogen species that continuously regenerate Brønsted acid sites on the HY zeolite to drive PCL protonation and chain scission. Concurrently, plasma-induced electric field polarization promotes electron redistribution toward the PCL ester groups, lowering the activation energy barrier from 2.29 to 1.71 eV and rendering γ-CL the kinetically and thermodynamically favored product. Notably, under the same reaction conditions at 200°C, the PCL conversion achieved by plasma catalysis is 8.77-fold higher than that obtained via thermal catalysis. Without solvent or external heating, this plasma approach achieves 100% PCL conversion with a 93.1% selectivity of caprolactones at a remarkably low temperature of 150°C. Finally, demonstrating macroscopic industrial viability, we successfully scaled the reactor 200-fold to process 100 g of PCL. By integrating a photovoltaic power system to eliminate grid reliance, this work establishes a decentralized, solar-driven paradigm for sustainable plastic upcycling.

2 | Results and Discussion

2.1 | Characterization of Synthesized M/HY Catalysts

Figure 2a illustrates the supported structure of the catalyst. The Si/Al ratio of M/HY and metal loading are listed in Table S1. This ion exchange-impregnation approach not only endows the Y zeolite surface with more Lewis acid sites but also introduces additional active metal sites on its surface, thereby enabling the catalyst to possess the capability of CH_4 dissociation. SEM and EDS mapping of Ni/HY (Figure 2b) reveal that the loaded Ni metal particles are uniformly distributed on HY, with agglomerated Ni particles predominantly ranging from 29 to 47 nm in size. From TEM results, it can be observed that the HY catalyst possesses pore structures with a pore diameter of approximately 0.6 nm, which is within the pore size range obtained from the Horvath-Kawazoe (HK) method as presented in Figure 2d. Despite a 5.0 wt% loading, Ni/HY shows no distinct metallic Ni XRD peaks (Figure 2c) because the ultra-small Ni precursors primarily migrate into the 0.46 nm micropores, as evidenced by the disappearance of these pores in BET analysis. According to the Scherrer equation, such sub-nanometer confinement ($<$ 1 nm) induces extreme peak broadening, resulting in their weak signals being completely masked by the intense FAU framework peaks. Figure 2d shows that M/HY catalysts exhibited composite type I/IV isotherms with H2 hysteresis loops, indicating their microporous structures with some presence of mesopores. The introduction of active metals has correspondingly taken up some of the pores of HY. The unloaded HY catalyst possesses a microporous structure with a most probable pore size of 0.46 nm (Table S2), which essentially disappears after the loading of active metals.

NH_3-TPD results (Figure 2e) indicate that both strong acid sites and weak acid sites coexist on the M/HY catalyst (at ∼200°C, corresponding to the desorption from weak acid sites) and high-temperature regions (at ∼400°C, corresponding to the desorption from strong acid sites) [30, 31]. Compared to HY, the loading of active metals resulted in a reduction in the NH_3 desorption amount, as presented in Table S3. This can be attributed to partial pore blockage and reduced specific surface area upon metal loading. Further, Py-IR was employed to quantitatively analyze the density of Brønsted and Lewis acid sites (Figure 2f). The absorption peaks located at 1450 and 1540 cm^{-1} correspond to the adsorption of the pyridine on Lewis and Brønsted acid sites, respectively, while the absorption signal at 1486 cm^{-1} arises from the overlapping contributions of pyridine adsorbed on both Lewis and Brønsted acid sites [32, 33]. As presented in Table S4, consistent with the reduction in total acid amount, after loading 5 wt% Ni, the densities of Brønsted and Lewis acid sites decreased from 0.51 and 0.22 mmol g^{-1} (HY) to 0.33 and 0.27 mmol g^{-1} (Ni/HY), respectively. The TGA-DSC results (Figure S2) indicated that the introduction of HY catalyst or metal-loaded HY catalysts significantly reduced the pyrolysis temperature of PCL compared with pyrolysis only.

2.2 | Diagnosis of Plasma During Plasma-Catalytic PCL Hydrogenolysis

Four gases (He, H_2, CH_4, and C_2H_4) with different hydrogen contents were selected as the plasma source gases. The Lissajous figures (Figure 3a) are all parallelograms, indicating a filamentary discharge mode of the plasma [34, 35]. Figure 3b presents the voltage-current (U-I) curves in the presence of PCL and Ni/HY. Using H_2 and CH_4 as plasma source gases achieved a more intense plasma discharge. This observation aligns with the characterization of plasma discharge properties via Lissajous figures (Table S5), where the C_{eff} measured 61.3 pF when using C_2H_4 as the plasma source gas, compared to 104.8 and 88.4 pF for H_2 and CH_4, respectively, suggesting a higher plasma density when using H_2 and CH_4 plasmas. The DBD reactor was operated at room temperature (Figure 3c), while the temperature of the reaction region was maintained by the waste heat generated during the plasma discharge to facilitate the melting of PCL, enabling it to access the active sites of HY and thereby overcome mass transfer limitations.

Compared to HY, the loading of active metal did not induce significant changes in the plasma discharge characteristics (Figures S3a,b); however, Ni loading enhanced the intensity of H· and CH· radical spectral lines (Table S6). As presented in Figure 3d, the formation of CH^+ and CH· radicals can be attributed to the electron-impact dissociation of CH_4, while the production of CO suggests that PCL was decomposed during the plasma discharge, where oxygen atoms from the polymer chains dissociate into oxygen radicals and further recombine with carbon radicals to form CO and CO_2 [36–38]. The appearance of the H spectral line at 656.3 nm indicates the generation of H radicals during CH_4 decomposition. Normalized spectral line intensities show that CH and H radicals significantly increased after the introduction of Ni, suggesting that Ni sites further promoted CH_4 cracking into H radicals via surface C─H bond scission, thereby enhancing the concentration of hydrogen radicals in the methane plasma [39]. The results suggest that the use of inexpensive CH_4 as a plasma source gas, combined with Ni/HY, achieved cracking of CH_4 to in situ supply hydrogen. Figure 3e shows the time-averaged spatial distributions of CH_3· and H· radicals, as well as the electric field distribution. The simulated reactions are listed in Table S7. CH_3· and H· radicals are primarily present near PCL-HY powders and around the electrodes, which suggests that the introduction of Ni/HY and PCL amplifies the local electric field, thereby facilitating CH_4 dissociation, increasing H· concentration on Ni/HY surface, and thus promoting PCL hydrogenolysis. Figure 3f shows that the TG curve of the residues exhibits a single weight-loss signal below 250°C, which can be attributed to the volatilization of the adsorbed products. To further identify the types of trace residual substances, we employed ATR–FTIR to analyze the functional groups present in the solid residues (Figure 3g). For the fresh PCL-Ni/HY mixture, the FTIR spectra show characteristic absorption bands corresponding to methylene stretching vibrations (2940 and 2860 cm^{-1}), ester carbonyl stretching (1720 cm^{-1}), and methylene bending vibrations (1457 cm^{-1}) [40]. These absorption signals disappeared in the residue after the plasma reaction, confirming that PCL was completely converted.

2.3 | Effect of Active Metal, Plasma Source Gas, and Operation Condition

Figure 4a illustrates the effect of the catalyst and the reaction conditions on the conversion of PCL and the formation of CL. The detailed product distributions are presented in Figures S5 and S6. To rigorously quantify the plasma synergy, a strictly temperature-matched control experiment was conducted. In conventional thermal-catalytic pyrolysis, PCL hardly undergoes decomposition at 200°C. However, at the identical bulk temperature of 200°C, the NTP-assisted process exhibited an 8.77-fold enhancement in the PCL conversion, firmly ruling out the effect of global Joule heating. Under these plasma conditions, the PCL conversion using the bare HY zeolite was 70.2%, with a CLs selectivity of 73.6%. The catalytic performance was further enhanced by metal modification, where Fe/HY and Ru/HY exhibited comparable PCL conversions and selectivities (81.2% and 89.4% for Fe/HY, and 83.9% and 82.3% for Ru/HY, respectively). Ultimately, the PCL conversion and CLs selectivity achieved using Ni/HY reached 100% and 92.1%, respectively, indicating that the introduction of Ni sites optimally promotes the dissociation of PCL and the generation of CL in the CH_4 plasma. The utilization of H_2 and CH_4 plasmas achieved complete PCL conversion (100%) with CLs selectivities of 84.3% and 92.1%, respectively, significantly outperforming those obtained under He plasma. Notably, when employing CH_4 as the feed gas, 14 vol% of H_2 was detected in the exhaust gas, demonstrating the feasibility of utilizing methane plasma as an in-situ hydrogen source (Figure S6). This phenomenon suggests that CH_4 underwent cracking into H radicals under the action of the Ni active sites. These H radicals can promote PCL decomposition through the hydrogenolysis reaction [41]. As the reaction progresses, PCL conversion increases but CLs selectivity slightly decreases (Figure 4b). Unlike purely thermal conditions, where the stable lactone ring resists secondary degradation at 150°C–200°C (93.7% of pure γ-CL remains intact after 1 h), the CH_4 plasma environment features highly energetic electrons. While these electrons profoundly accelerate primary PCL dissociation, prolonged exposure inevitably induces electron-impact

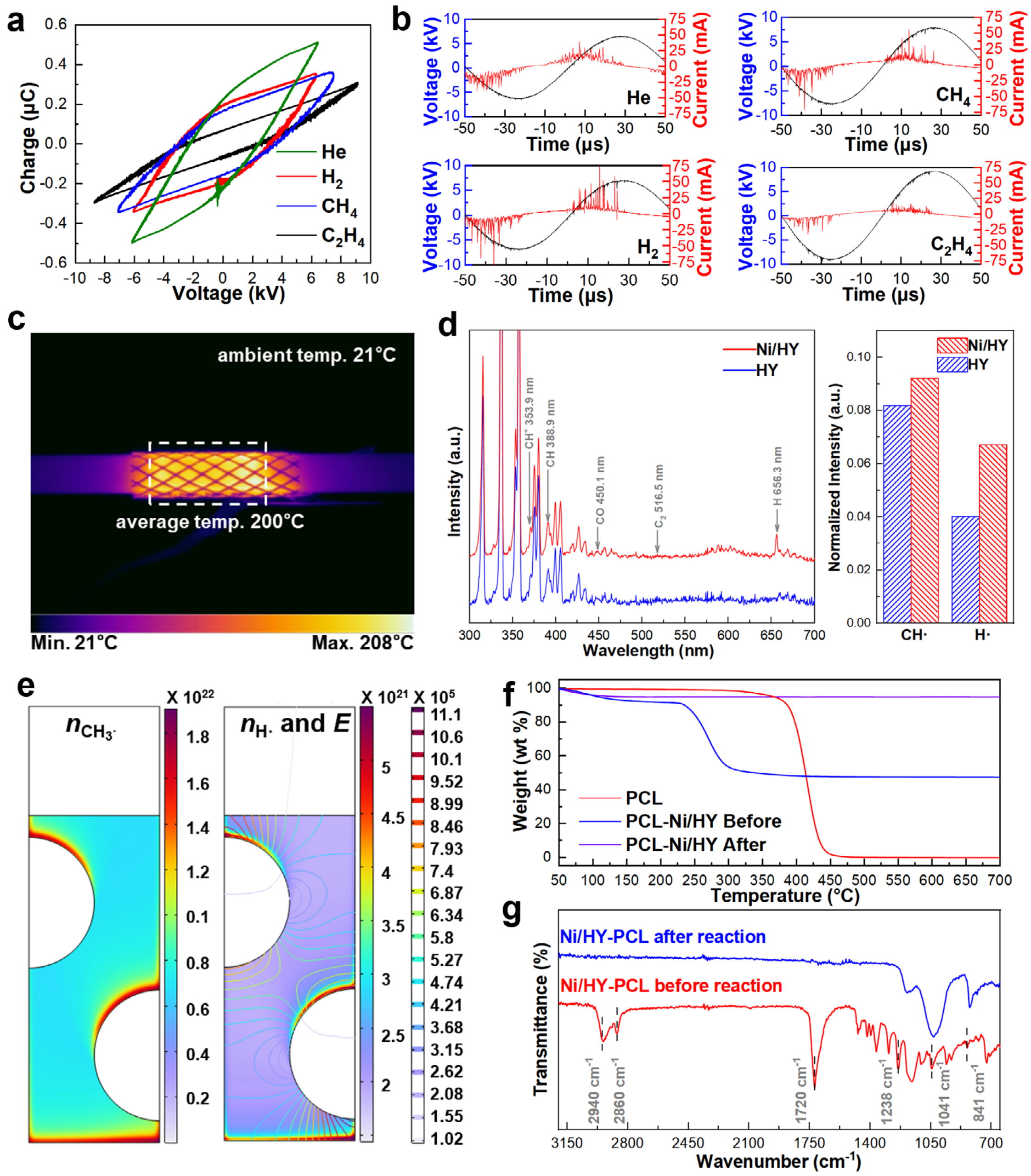

FIGURE 3 | Electro- and optical diagnosis of plasma-catalytic PCL hydrogenolysis process. (a) Lissajous figure using Ni/HY catalyst with different plasma source gases. (b) U-I curves using Ni/HY catalyst with different plasma source gases. (c) Temperature distribution using Ni/HY catalyst with CH_4 as plasma source gas at the plasma discharge power of 17 W. (d) OES spectra using Ni/HY and HY catalysts and CH_4 as plasma source gas representing the relative concentration of free radicals during PCL hydrogenolysis. (e) Simulation of CH_3 and H concentrations and electric field for CH_4 plasma discharge during a half-cycle of applied voltage. (f) TG spectra for PCL-Ni/HY mixture after 1 h CH_4 NTP discharge. (g) FTIR spectra for comparison of PCL-Ni/HY mixture before and after CH_4 NTP discharge for 1 h.

secondary reactions, causing the over-cracking of accumulated caprolactone. This is corroborated by control experiments showing that pure γ-CL survival drops to 84.6% upon CH_4 plasma exposure. Therefore, the progressive decline in selectivity is a natural kinetic consequence of electron-impact secondary degradation inherent to NTP systems. Further optimization of reaction duration and power (Figure 4c) demonstrated that the robust plasma activation allowed the complete decomposition of PCL to be achieved at a significantly reduced temperature of 150°C within a 3-hour reaction timeframe, achieving a CLs yield of 93.1%.

As shown in Figure 4d, the pure thermal controls (PCL-SiO_2-Therm, PCL-HY-Therm, and PCL-Ni/HY-Therm) exhibited negligible PCL conversion (below 14%) and minimal caprolactone yield, demonstrating that PCL decomposition is severely hindered under purely thermal conditions at 200°C. To further decouple plasma chemistry from generic Joule heating, CH_4 and H_2 plasmas were compared under identical conditions. Despite generating comparable macroscopic heat, the CH_4 plasma achieved remarkably higher γ-CL (88.9% vs. 79.7%). This discrepancy occurs because H_2 NTP exacerbates the reverse decomposition of CL products (Figure S7), thereby confirming

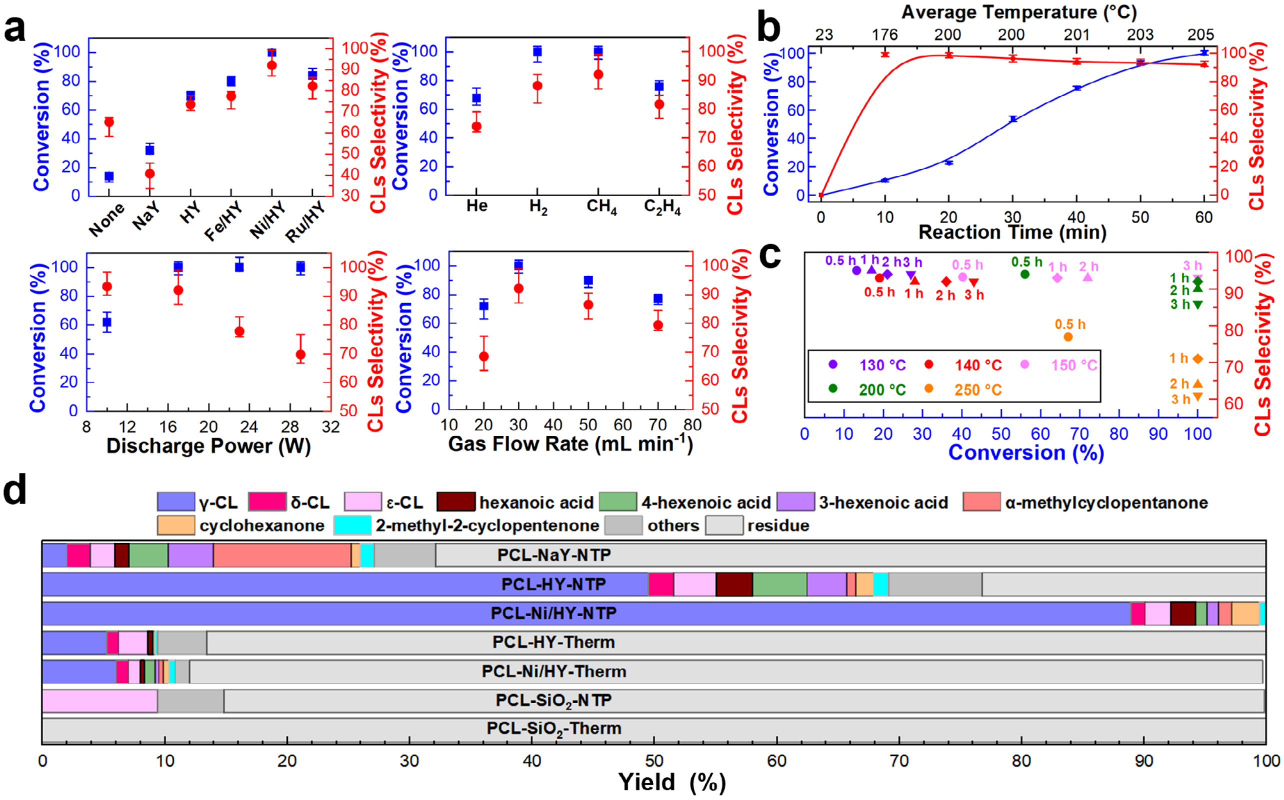

FIGURE 4 | PCL hydrogenolysis performance under different catalysts and operation conditions. (a) Effect of catalyst, plasma source gas, discharge power, and gas flow rate on PCL conversion and CL monomer after 1 h reaction. For the catalyst-free condition, the discharge region was packed with PCL and quartz sand. Plasma source gas, catalyst, plasma discharge power, and gas flow rate were CH_4, Ni/HY, 17 W (200°C), and 30 mL min^{-1}, respectively, unless otherwise specified. The temperatures of the discharge region were room temperature, 150°C, 170°C, 200°C, 230°C, and 260°C at 0, 7, 10, 17, 23, and 29 W, respectively, as presented in Figure S4. (b) Relationship of PCL conversion and CL generation with reaction time and temperature. (c) Optimization of reaction time and temperature. d Liquid products distribution after 1 h reaction when using CH_4 as plasma source gas at 30 mL min^{-1} and 200°C (the labels "Therm" and "NTP" denote thermal catalysis and plasma catalysis, respectively. "SiO_2, NaY, HY, and Ni/HY" denote the catalysts used under different conditions).

that specific plasma properties (radicals and EEDF) strictly dictate product, overriding any generic thermal effects. In the plasma-catalysis processes, although the presence of NaY (PCL-NaY-NTP) and HY (PCL-HY-NTP) promoted the depolymerization of PCL to C_6 ketones, C_6 acids (as determined by mass spectra in Figure S8), HY achieved a 51.7% yield of CL (γ-CL 46.2%, δ-CL 2.1%, and ε-CL 3.4%). Additionally, approximately 14% of by-products were identified, primarily consisting of C_6 acids and aldehydes. For PCL-Ni/HY-NTP, the yield of CL reached 92.1% (γ-CL 88.9%, δ-CL 1.1%, and ε-CL 2.1%). The exceptionally high yield toward γ-CL also effectively rules out the dominance of localized intense thermal spikes in the DBD plasma, which would otherwise induce unselective thermal cracking and severe gasification. As presented in Figure S5e, a 10-fold scale-up experiment achieved a 91.7% CLs yield. We further evaluated the catalytic performance of the regenerated Ni/HY, which still achieved 100% PCL conversion, along with 90.9% CLs yield and a γ-CL yield of 87.4%. Furthermore, to evaluate the stability of the active sites and the extent of carbon deposition during the plasma-catalytic process, comprehensive post-reaction characterizations were conducted on the spent Ni/HY catalyst. XPS (Figure S9a,c) analysis reveals that the active metallic Ni^0 state is robustly preserved after the reaction, indicating that the CH_4 plasma environment effectively prevents the severe oxidation of Ni sites by oxygen-containing fragments. TGA of the spent catalyst (Figure S9g) demonstrates that the carbon deposition is relatively mild (quantified at 4 wt%), confirming that the plasma-catalysis system is highly resistant to severe coking.

2.4 | Mechanism of CH_4 Plasma-Catalytic Hydrogenolysis of PCL

To investigate the role of CH_4 plasma in the hydrogenolysis of PCL over Ni/HY, we conducted CD_4 and D_2 isotope labeling experiments to study the transfer pathways of free radicals in the plasma. Figure 5a shows the mass spectrum of γ-CL. When using CH_4-CD_4 mixture as plasma source gas, new fragment signals appeared at: m/z = 75 and 76, which can be attributed to the formation of $C_3H_5DO_2$ and $C_3H_4D_2O_2$; m/z = 87 and 88, corresponding to $C_4H_5DO_2$ and $C_4H_4D_2O_2$; m/z = 115 and 116, derived from $C_6H_9DO_2$ and $C_6H_8D_2O_2$. These species originate from the replacement of H atoms on CLs by D generated from CD_4 cracking, as revealed by ^{1}H-NMR in Figure S10.

As shown in Figure 5b, when using the deuterium-labeled catalyst (Ni/HY-d) under normal CH_4 plasma, the mass spectrum of the obtained γ-CL closely resembled that from the CH_4/CD_4

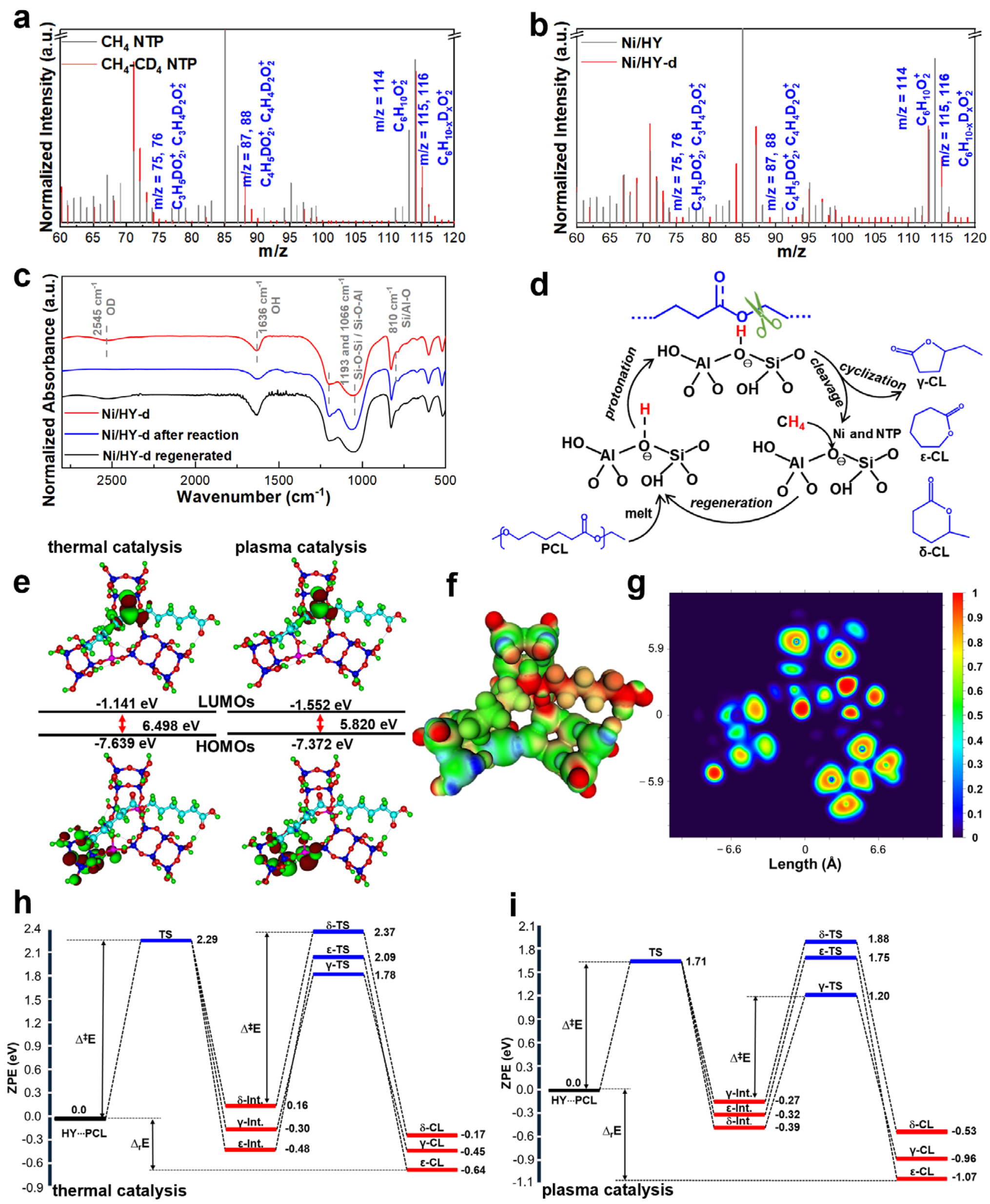

FIGURE 5 | Characterization of H transfer route during CH_4 NTP catalytic hydrogenolysis of PCL. (a) Mass spectra γ-CL produced using CH_4 and CH_4-CD_4 as plasma source gases. (b) Mass spectra γ-CL monomer produced from CH_4 and CH_4-CD_4 plasma discharge. (c) FTIR spectra of ^{2}H-labeled Ni/HY (Ni/HY-d) before and after reaction using CH_4 NTP. (d) Proposed H transfer route between PCL and Ni/HY. (e) Frontier molecular orbitals (HOMO/LUMO) and frontier orbitals under plasma conditions, showing LUMO stabilization. (f) Charge density difference map highlighting polarization effects. Red regions indicate electron accumulation, blue regions indicate electron depletion, and green regions represent neutral areas. g Electron localization function (ELF) plot demonstrating field distribution. (h, i) Calculated reaction coordinate for (h) thermal-catalytic and (i) plasma-catalytic hydrogenolysis of PCL showing protonation, C—O cleavage, and ring formation pathways with corresponding energy barriers.

NTP experiment, showing signals corresponding to deuterated γ-CL. Corroborating this, FTIR spectra (Figure 5c) of the Ni/HY-d catalyst exhibited an infrared absorption peak at 2545 cm^{-1}, attributed to the O—D stretching vibration [42], which completely disappeared after 1 h of reaction. These observations confirm the occurrence of hydrogen transfer from Ni/HY to γ-CL during PCL hydrogenolysis. Furthermore, structural analysis (Figure S10) reveals that the transferred deuterium is primarily localized at the α-methylene position of the lactone ring, driven by the acid-catalyzed hydrogenolysis. The post-reaction Ni/HY catalyst was further treated with D_2 plasma. The regeneration of O—D groups suggests that the D_2 plasma treatment successfully reconstructed the O—D sites on the Ni/HY. The results indicate a dynamic consumption and reconstruction process of O—H groups on the HY during plasma-catalytic PCL hydrogenolysis. Specifically, the hydrogen from HY participates in PCL hydrogenolysis, while the hydrogen supplied by the CH_4 plasma sustains the process by replenishing the O—H sites. As shown in Figure 5d, when PCL chains adsorb onto the acidic sites of the Ni/HY surface, these sites act as hydrogen donors to protonate PCL, facilitating chain scission. The cleaved fragments desorb from active sites for further dissociation and cyclization into γ-CL, while the protondepleted sites are regenerated by capturing H^+ and H· from the CH_4 plasma. These atomic H species can originate either from the diffusion of H species in the gas phase or from the secondary dissociation and subsequent diffusion of CH_x fragments (formed by the partial dissociation of CH_4) on the Ni sites. Coupled with the observation of O─D bond regeneration, these findings construct a closed-loop H transfer mechanism: plasma-generated active species regenerate surface Brønsted acid sites, which subsequently mediate the depolymerization and cyclization of PCL.

To further elucidate the reaction mechanisms of PCL cleavage and cyclization, DFT calculations were performed under thermal- and plasma-catalysis conditions for comparison. For thermal-catalytic hydrogenolysis (Figure 5e), the HOMO–LUMO gap was calculated to be 6.498 eV, which decreased to 5.820 eV under plasma-catalytic hydrogenolysis. This notable reduction of 0.678 eV is primarily attributed to a pronounced stabilization of the LUMO energy from −1.141 to −1.552 eV, while the HOMO energy remains nearly unchanged. The HOMO is primarily localized on the HY, indicating that the HY contributes electron density available for PCL cleavage. In contrast, the LUMO is more delocalized, extending over the antibonding σ*(C─O) and π*(C═O) orbitals of PCL, positioned directly above the Brønsted acid site, suggesting that electron donation from the HY to the electron-deficient sites of PCL is facilitated, thereby enhancing adsorption and activating PCL toward subsequent bond cleavage reactions [43].

The charge density difference plot (Figure 5f) of the CL dimer on HY reveals a pronounced localization of electron density after adsorption. The intense red zone around the central carbonyl region of PCL indicates a high electron density due to the electronegative oxygen and π-bonding in the C═O group, suggesting a strong nucleophilic feature that facilitates interaction with the acidic proton of HY [44]. This charge accumulation supports the protonation and C─O bond cleavage, where the electron-rich carbonyl oxygen acts as the primary site for proton donation from Al─O(H)─Si, stabilizing the adsorption complex. The PCL-HY interface shows strong localization (ELF > 0.8) at carbonyl and ether oxygens, confirming lone pairs crucial for protonation (Figure 5g), while moderate ELF values (0.4–0.6) suggest physisorption of PCL, which aligns with adsorption energy trends and suggests that thermal catalysis (−0.16 eV) disrupts and plasma catalysis (−0.41 eV) further modulates interactions through dipole reorientation [45, 46]. This analysis indicates that PCL adsorption on HY is driven by acid-base interactions between them and enhanced by plasma-induced electric field polarization. As presented in Figure S11, the adsorption energy (ΔG_{ads}) of CL dimer shows a trend of −0.607 eV at room temperature, −0.157 eV for thermal catalysis, −0.77 eV under EF alone, and −0.41 eV for plasma catalysis. The strongest adsorption under EF alone reflects enhanced H-bonding or electrostatic interactions due to polarization of polar groups with the Brønsted acid site of HY under electric field polarization, surpassing the −0.607 eV at RT driven by enthalpic gains [47]. At 200°C, ΔG_{ads} drops to −0.15 eV due to entropic penalties ($-T\Delta S_{ads}$), and further to −0.41 eV for plasma catalysis, where plasma-generated thermal disruption and electric field-induced dipole misalignment or repulsion weaken the interaction [48].

Before mapping the reaction profiles, we evaluated four aluminum substitution positions in the HY framework, exclusively selecting the most thermodynamically stable configuration (HY-1) for all subsequent mechanistic calculations (Figure S12). Figures 5h,i present the reaction coordinate profile revealing the conversion mechanism of PCL. The geometry configurations of intermediates and transition states are presented in Figure S13. In both thermal and plasma catalysis, the reaction is initiated by a concerted step involving protonation of the carbonyl oxygen and C─O bond cleavage, yielding the ε-Int, with γ-Int and δ-Int also formed via intramolecular proton transfer or proton abstraction by plasma radicals when conducting plasma catalysis. The energetic requirements for this step differ substantially: the barrier is 2.29 eV for thermal catalysis, whereas plasma catalysis reduces it to 1.71 eV, underscoring the role of plasma in facilitating the cleavage of PCL and explaining the higher PCL conversion rate during plasma catalysis. The intermediates lead to distinct CLs. For thermal catalysis at 200°C, the generation of γ-CL proceeds through an intermediate at −0.30 eV and requires a relative activation energy barrier of 2.08 eV, yielding the γ-CL at −0.45 eV. The δ-pathway is less favorable, with a higher barrier of 2.21 eV, and modest stabilization of the δ-CL (−0.17 eV). The generation of ε-CL furnishes a thermodynamically stable product (−0.64 eV), but requires a substantial barrier of 2.57 eV, limiting kinetic accessibility. The barrier of the generation of γ-CL drops to 1.47 eV for the case of plasma catalysis, affording a γ-CL at −0.96 eV, with exothermicity increasing by 0.51 eV. The reduction in reaction barriers can be attributed to the fact that plasma discharge induces polarization of the C─O and C═O bonds in the PCL, thereby stabilizing charge-separated transition states and favoring the compact five-membered-ring geometry of γ-CL. This effect manifests as a significant barrier reduction of 0.61 eV relative to the thermal catalysis. By contrast, the ε-pathway suffers from a comparatively high barrier of 2.07 eV, while the δ-pathway remains the least favorable due to both its elevated barrier (2.27 eV) and modest stabilization (−0.53 eV). The plasma also increases exothermicity of the products, yet the γ-pathway's kinetic advantage, driven by a significant barrier reduction (0.58 eV), ensures the predominance of γ-CL generation. The substantially lower activation barrier for the γ-CL pathway translates to a reaction rate that is orders of magnitude higher than those of the competing pathways. This profound kinetic divergence effectively shuts down any competitive cyclization. Furthermore, while this idealized projection dictates exclusive γ-CL formation, the inherent macroscopic heterogeneity of the plasma discharge in the catalyst-packed region explains the experimental 93.1% yield, alongside trace formations of ε-CL and δ-CL. This confirms that plasma-induced dipole reorientation unequivocally secures γ-CL as the overwhelmingly dominant product. These findings definitively prove the nonthermal plasma-catalytic synergy. Unlike generic heat, which only helps overcome existing barriers, the plasma-induced electric field fundamentally lowers the C─O cleavage barrier (from 2.29 to 1.71 eV). Corroborating this, unconstrained geometric optimization confirms that this radical-induced hydrogen abstraction proceeds as a spontaneous process, creating the essential reactive pivot for subsequent γ-CL formation (Figure S14). To ensure the robustness of these mechanistic insights, potential energy surface evaluations across varying electric field strengths confirm a monotonic decrease in activation barriers, firmly substantiating this field-induced catalytic enhancement (Figure S15). This firmly establishes the

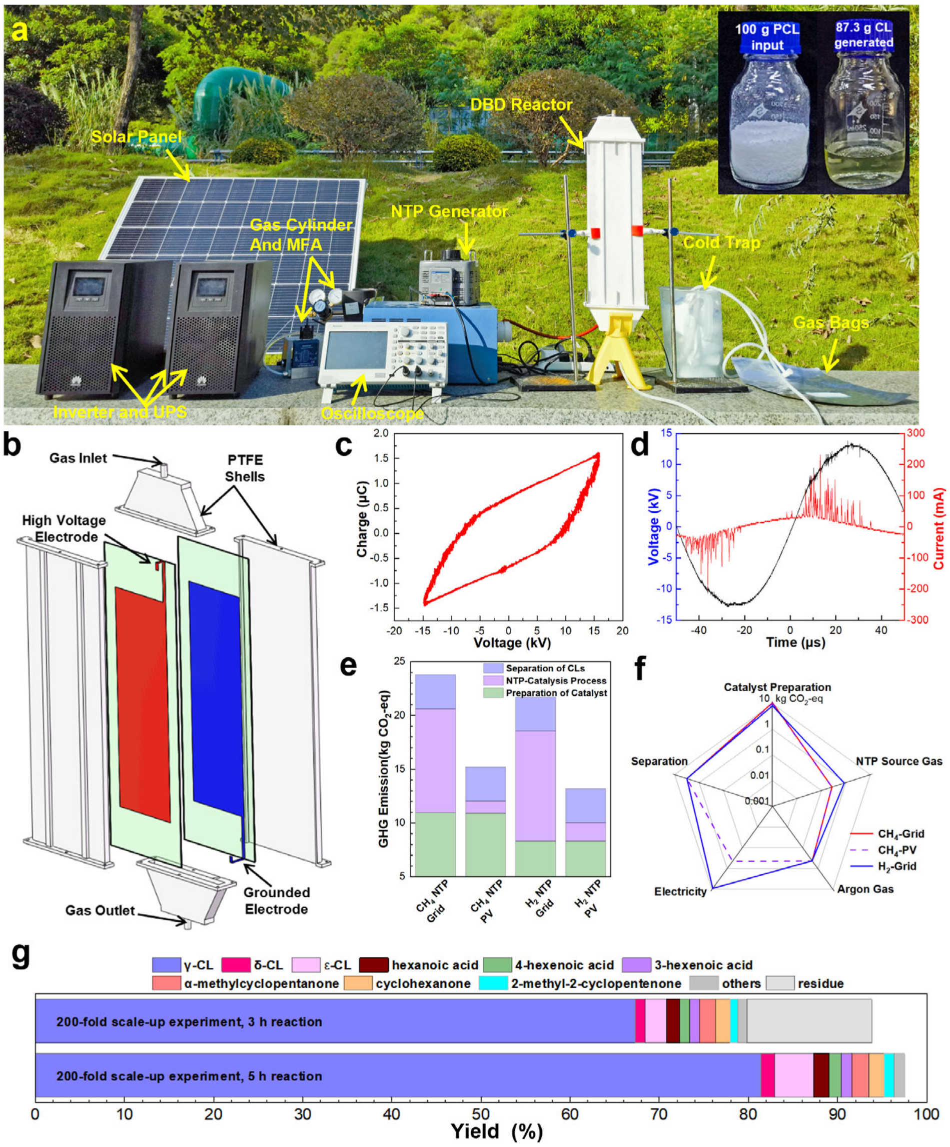

FIGURE 6 | 200-fold scale-up experiment. (a) Experimental set-up powered by solar energy; (b) exploded view of the DBD reactor. The reactor has a double-layer dielectric plate structure, with a dielectric layer thickness of 1 mm and a discharge gap of 9 mm. The discharge area can be adjusted by varying the electrode area, and in this work, the plasma discharge area was fixed at 0.02 m^2; (c, d) Lissajous figure and U-I curve of plasma discharge; (e, f) GHG emissions in different reaction conditions; (g) product distribution for scaled-up experiments.

plasma as a synergistic reactant and field-promoter, rather than a localized heating source.

2.5 | Experiment Scale-Up and Life Cycle Assessment

As presented in Figure 6a, we constructed a photovoltaic (PV)-powered plasma-catalytic hydrogenolysis demonstration unit for PCL, which can operate entirely on solar energy under suitable climatic conditions. The core of the system is a flat-plate dual-layer DBD reactor (Figure 6b) with adjustable discharge area and gap, enabling flexible treatment capacities. As shown in Figure 6c,d, the plasma discharge exhibits a filamentary discharge mode with a larger C_{eff} (243 pF) and Q_{trans} (1.63 μC), confirming the realization of large-area uniform discharge.

Life cycle assessment (LCA) was conducted to evaluate greenhouse gas (GHG) emissions during PCL hydrogenolysis (Figure 6e,f). The corresponding life cycle inventory is presented in Table S8. The main GHG sources are electricity use and upstream chemical production emissions from three stages: catalyst preparation, plasma reaction, and product separation. Using CH_4 results in 2.6 $kgCO_2$-eq higher GHG emissions than H_2 due to the loading of Ni in catalyst preparation stage. In the plasma reaction stage, CH_4 as the source gas emits 0.59 $kgCO_2$-eq less GHG than H_2. Product separation emissions mainly come from rotary evaporation, electricity use, and solvent loss. Applying PV power in the plasma process reduces GHG emissions from 9.65 $kgCO_2$-eq to 1.13 $kgCO_2$-eq, mainly from source gas use and potential emissions from PV power station operations. With PV-powered CH_4 plasma, the core conversion stage of PCL hydrogenolysis emits only 1.13 $kgCO_2$-eq, 34% lower than PV-powered H_2 plasma. To contextualize the calculated carbon footprint (1.13 kg CO_2-eq kg^{-1} PCL), we compared our result against two conventional benchmarks: (i) the direct incineration of PCL waste (~2.3–2.5 kg CO_2-eq kg^{-1}) and (ii) the virgin production of ε-CL from fossil resources (~4.0–8.0 kg CO_2-eq kg^{-1}) [49, 50]. This comparison confirms that our plasma-catalytic strategy offers a substantially more sustainable pathway for plastic circularity. This sustainable profile is further reinforced by the exceptional durability of the Ni/HY catalyst, which maintains robust PCL conversion and high γ-CL yield over five consecutive regeneration cycles (Figure S16a)

The successful 200-fold scale-up demonstration, processing 100 g of PCL in a single batch, validates the potential of this plasma-catalytic strategy for larger-scale plastic chemical upcycling. However, this scale-up also revealed challenges, manifesting as a longer reaction time required for complete conversion (5 vs. 3 h) and a moderate decrease in the CLs yield (87.3%) compared to the optimized small-scale reactions (Figure 6g). This is attributed to challenges in maintaining uniform plasma conditions in a larger reactor volume. Less efficient distribution of H radicals likely slowed hydrogenolysis and promoted minor side reactions. These results highlight that scaling plasma processes requires managing transport phenomena. This is highly consistent with our catalyst dosage evaluations, which demonstrated that suboptimal catalyst-to-plastic ratios induce severe mass-transfer barriers within the polymer melt, drastically inhibiting conversion (Figure S16b). Future work will optimize reactor design, such as employing multi-point electrodes or fluidized beds, to enhance plasma uniformity and radical-catalyst contact for improved performance at industrial scales.

3 | Conclusion

In this work, we utilized CH_4 plasma over a Ni/HY catalyst to synergistically couple CH_4 cracking with PCL depolymerization, thereby enabling solvent-free hydrogenolysis driven by in situ generated hydrogen radicals. This strategy validates the feasibility of replacing high-pressure H_2 and noble-metal catalysts required in conventional hydrogenolysis with ambient-pressure CH_4 NTP and Ni-based catalyst, thereby improving process sustainability. Fundamentally, this superior plasma-catalytic performance is strictly governed by three synergistic mechanisms. First, the highly dispersed Ni sites maximize the in situ generation of hydrogen radicals without triggering aggressive secondary over-cracking. Second, the HY zeolite framework provides a robust Brønsted acid density to dynamically capture these radicals and sustain continuous PCL protonation. Finally, the plasma-induced orbital polarization provides the critical electronic modulation to steer the highly selective cyclization pathway. By significantly stabilizing the LUMO energy, this localized electric field uniquely lowers the primary PCL dissociation barrier and sterically steers the cyclization toward γ-CL. Consequently, we achieved 100% PCL conversion and a remarkable 93.1% yield of caprolactones at a low temperature of 150°C. Furthermore, bridging the critical translational gap from milligram-scale analytical studies to practical engineering, we successfully executed a 200-fold scale-up demonstration. The custom-designed, flat-plate double-dielectric-layer DBD reactor allowed for the continuous processing of a 100 g batch of PCL. Powered entirely off-grid via an integrated photovoltaic system, this work not only elucidates a rigorous structure–performance relationship for ester bond activation but also provides a tangible, decentralized, and highly scalable technological blueprint for sustainable plastic upcycling. Far outperforming conventional thermochemical methods, this plasma-catalytic hydrogenolysis enables the highly selective upcycling of PCL with unprecedented efficiency and sustainability.

Author Contributions

Jin Liu: methodology, software, data curation, formal analysis, investigation, visualization, writing – original draft, writing - review and editing, validation. **Ayyaz Mahmood**: methodology, investigation, formal analysis, writing – original draft, writing – review and editing, visualization, data curation, validation. **Sijie Li**: methodology, formal analysis, resources, validation. **Yuan Xue**: methodology, writing – review and editing. **Mo Zheng**: methodology, writing – review and editing, software. **Yin-Ning Zhou**: writing – review and editing. **You-Wei Cheng**: writing – review and editing. **Xi Gao**: conceptualization, methodology, supervision, writing – original draft, writing – review and editing, funding acquisition, project administration, software. **Zheng-Hong Luo**: writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

The authors have cited additional references within the Supporting Information [51–84].

Supporting File: anie74323-sup-0001-SuppMat.docx.